



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **AUC:** Area Under the (ROC) Curve
- ▣ **BERT:** Bidirectional Encoder Representations from Transformers
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CAR:** Cumulative Abnormal Return
- ▣ **CI:** Confidence Interval
- ▣ **ECE:** Expected Calibration Error
- ▣ **FNSPID:** Financial News and Stock Price Integration Dataset
- ▣ **GPT-4:** Generative Pre-trained Transformer 4
- ▣ **LLM:** Large Language Model
- ▣ **LM:** Loughran-McDonald (financial sentiment dictionary)
- ▣ **NLL:** Negative Log-Likelihood
- ▣ **OLS:** Ordinary Least Squares
- ▣ **PPI:** Prediction-Powered Inference
- ▣ **ROC:** Receiver Operating Characteristic (curve)
- ▣ **SD:** Standard Deviation
- ▣ **SE:** Standard Error
- ▣ **TF-IDF:** Term Frequency - Inverse Document Frequency
- ▣ **US:** United States
- ▣ **UTC:** Coordinated Universal Time (Temps Universel Coordonné)

Today's Questions and Seminar Structure

- ▣ Questions of the seminar
 - ▶ **can a language model read financial sentiment?**
 - ▶ **does that sentiment predict returns one can still trade?**
- ▣ Outputs to produce
 - ▶ the answers to the A exercises
 - ▶ one table and one figure per B exercise
 - ▶ the C1 corrections
 - ▶ one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#)
 - ▶ each task names its notebook section
- ▣ Exercises, types and order: the Exercise Map (next slide)
- ▣ Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how much does a misclassified dummy shrink a slope?	Solved	–
A2	how precise, and how powerful, is a paired accuracy test?	Solved	–
A3	at what cost per trade does the strategy stop earning?	Solved	–
B1	is FinBERT more accurate than the Loughran–McDonald (LM) dictionary?	Solved	–
B2	do headlines predict returns, or describe them?	Solved	–
B3	does the LLM remember the market?	Solved	–
C1	what is wrong in an AI answer?	Proposed	B1, B2, B3
B4	is a 7-billion-parameter LLM better than FinBERT? (homework)	Proposed	B1
Extension packs	A4–A8, B5–B11, C2, C3 (optional, after the lecture)	both	named on each task

- ☐ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ☐ Part A: derivations on paper; Part B: verification and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Used

Data	Content	Saved scores	Used in
Twitter Financial News (validation set, Hugging Face Hub)	2 388 headlines from 2022: 1 566 neutral, 475 positive, 347 negative	LM, FinBERT, Qwen2.5 (P1–P3), MiniLM	A1, A2, A4, A6–A8, B1, B4–B8, C1
Financial PhraseBank Malo et al. (2014)	4 846 news sentences, 2 264 with full agreement	same scores	B5
FNSPID Dong, Fan & Peng (2024)	106 820 headlines on 16 US stocks, 27.10.2009–18.12.2023	daily LM, FinBERT, Qwen2.5-7B scores	A3, A8, B2, B9–B11, C1–C3
Stocks and SPY	adjusted daily prices of the course data, one series per asset (field used: adjusted close), 2009–2023	–	B2, B11, C2, C3
S&P 500	monthly closes, Jan 2000 – Aug 2026	Qwen2.5 answers about each month	B3, C1

- The scores are saved with the chapter's Quantlets, [Quantlets/Ch_15](#): no language model has to run in class
- C3 (extension pack) needs a new scoring run of the FNSPID headlines

Setup: Run and Verify

```
# notebook section "Setup" defines the functions that
# read the data
pb, tw = eval_texts()    # PhraseBank, Twitter
P, rets = news_panel()   # FNSPID stock-days
print(len(pb), len(tw))
print(tw['label'].value_counts())
print(P.shape,
      P.index.get_level_values('ticker').nunique())
```

- Expected output
 - ▶ 4 846 and 2 388 rows
 - ▶ neutral 1 566, positive 475, negative 347
 - ▶ (18 282, 19) and 16 stocks
- Each row of tw holds the text, the human label and the probabilities of every model (e.g. finbert_negative, qwen7B_positive)
- If your counts differ, stop and check the files before any exercise

Notation Used in Every Exercise

- Dictionary tone τ and the Loughran–McDonald (LM) lists [Loughran & McDonald \(2011\)](#)
 - ▶ definition and example: [What You Need \(1/20\)](#)
- Model probabilities p_k , label $\hat{y}$ and net score $s = p_{\text{pos}} - p_{\text{neg}}$ (FinBERT [Araci \(2019\)](#), Qwen2.5 [Qwen Team \(2024\)](#))
 - ▶ softmax with temperature T : [What You Need \(2/20\)](#)
- Two classifiers A and B on the same texts: discordant counts n_{01} , n_{10} and the McNemar test [McNemar \(1947\)](#)
 - ▶ definitions and the exact test: [What You Need \(8/20\)](#)
- Day d , horizon h and the first tradable return $d + 2$
 - ▶ timing of a headline: [What You Need \(13/20\)](#)
- Units and inference
 - ▶ bp: basis point, 0.01%; SE: standard error; CI: 95% confidence interval
 - ▶ t_{NW} : Newey–West t [Newey & West \(1987\)](#), [What You Need \(16/20\)](#); clustered SE [Petersen \(2009\)](#), [What You Need \(14/20\)](#)

What You Need for Today (1/20): Three Ways to Read Sentiment

- **Dictionary** (bag of words) [Loughran & McDonald \(2011\)](#): the tone of a text

$$\tau = \frac{P - N}{P + N} \in [-1, 1]$$

- ▶ P, N : numbers of words of the text found in a fixed positive and a fixed negative list
- ▶ Class = sign of τ ($\tau = 0$: neutral); example $P = 2, N = 1$: $\tau = 1/3$, positive
- ▶ LM lists, built from 10-K filings: 2 355 negative and only 354 positive words; transparent, but blind to context and negation

- **FinBERT** [Araci \(2019\)](#): BERT, a Transformer encoder ([Chapter 14](#)), trained further on financial news

- ▶ Fine-tuned on labelled sentences; output: $p_{\text{pos}}, p_{\text{neu}}, p_{\text{neg}}$, summing to 1
- ▶ Fine-tuned on Financial PhraseBank [Malo et al. \(2014\)](#): its accuracy there is partly in-sample

- **LLM zero-shot** (Qwen2.5 [Qwen Team \(2024\)](#)): a large language model asked “Is this headline positive, negative or neutral?”

- ▶ No training on these labels (zero-shot); we read the logits z_k of the three answer words (next slide)

What You Need for Today (2/20): From Logits to Probabilities

- Logit z_k : the model's unnormalised score of answer $k \in \{\text{pos}, \text{neu}, \text{neg}\}$; larger means more likely
- **Softmax with temperature** $T > 0$: turns logits into probabilities

$$p_k(T) = \frac{e^{z_k/T}}{\sum_j e^{z_j/T}}, \quad \sum_k p_k(T) = 1$$

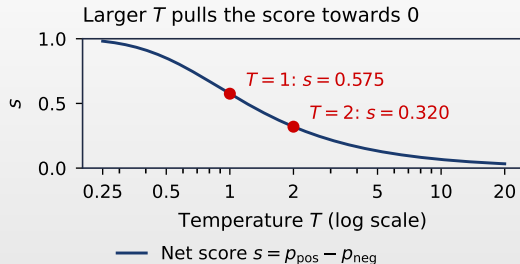
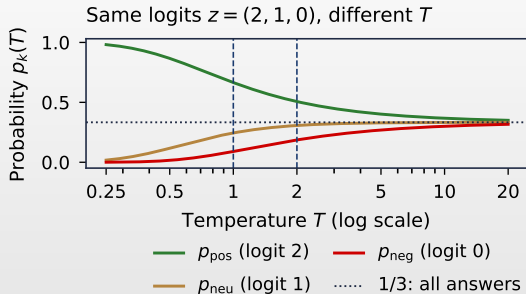
- ▶ $T = 1$: the model's own probabilities; $T > 1$ flattens them towards $1/3$; $T < 1$ sharpens them
- ▶ Only differences of logits matter: $z_k - z_j = T \ln(p_k/p_j)$, the logit gap (A7)

- **Label and net score**

$$\hat{y} = \arg \max_k p_k, \quad s = p_{\text{pos}} - p_{\text{neg}} \in [-1, 1]$$

- ▶ $\arg \max_k$: the answer with the largest probability; T does not change it, since dividing by $T > 0$ keeps the order of the logits
- ▶ $s > 0$: positive tone; s near 0: neutral or undecided; s changes with T

What You Need for Today (3/20): The Temperature at Work



- Example: logits $z = (2, 1, 0)$ for (pos, neu, neg)
 - ▶ $T = 1$: $p = (0.665; 0.245; 0.090)$, $s = 0.575$
 - ▶ $T = 2$: $p = (0.506; 0.307; 0.186)$, $s = 0.320$: same label, score pulled towards 0
- As $T \rightarrow \infty$ all three probabilities tend to $1/3$ and $s \rightarrow 0$; a threshold rule on s therefore depends on T

What You Need for Today (4/20): Scoring a Classifier

- Confusion matrix C : C_{jk} = number of texts with human label j (row, the truth) and predicted label k (column); n texts, K classes
- **Accuracy**: share of texts on the diagonal

$$\text{acc} = \frac{1}{n} \sum_{k=1}^K C_{kk}$$

- ▶ Benchmark: the share of the majority class (always predict it), never $1/K$

- **Precision** and **recall** of class k

$$\text{prec}_k = \frac{C_{kk}}{\sum_j C_{jk}}, \quad \text{rec}_k = \frac{C_{kk}}{\sum_j C_{kj}}$$

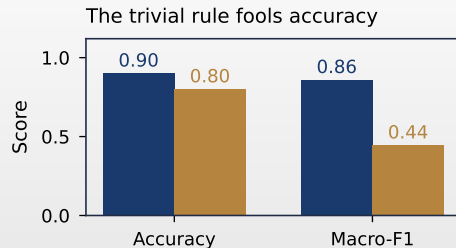
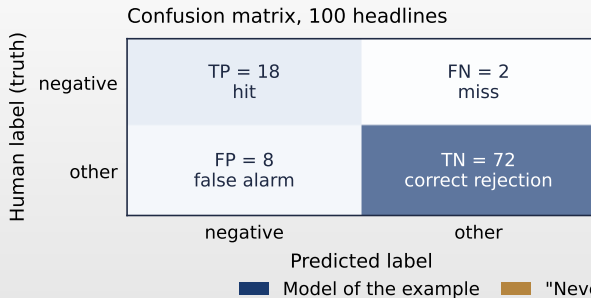
- ▶ prec: share of the texts flagged k that truly are k (column); rec: share of the true k that are flagged (row)

- F_1 and **macro-F1**

$$F_{1,k} = \frac{2 \text{prec}_k \text{rec}_k}{\text{prec}_k + \text{rec}_k}, \quad \text{macro-F1} = \frac{1}{K} \sum_{k=1}^K F_{1,k}$$

- ▶ F_1 : harmonic mean, low if either is low; macro-F1 gives a rare class the same weight as a frequent one

What You Need for Today (5/20): Accuracy, Macro-F1 and the Majority Class



- Example: 100 headlines, 20 truly negative; the model flags 18 of them and 8 of the other 80
 - ▶ Accuracy $(18 + 72)/100 = 0.90$, but "never negative" already scores 0.80; macro-F1: $(0.783 + 0.935)/2 = 0.86$ against $(0 + 0.889)/2 = 0.44$
 - ▶ Rates used in A1: $\pi = 0.20$ truly negative, $p = 0.26$ flagged, misses $\alpha_1 = 2/20 = 0.10$, false alarms $\alpha_0 = 8/80 = 0.10$

What You Need for Today (6/20): Measurement Error and Attenuation

- OLS slope on one regressor x : in large samples a ratio of moments

$$\text{plim}_{n \rightarrow \infty} \hat{b} = \frac{\text{Cov}(r, x)}{\text{Var}(x)}$$

- ▶ plim: limit in probability; r : the outcome (a return)

- **Classical** error: we observe $x = x^* + u$, u independent of x^* and of ε

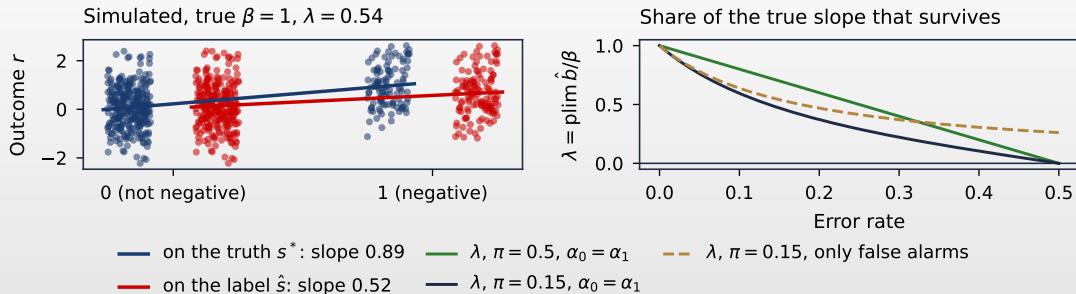
$$r = a + \beta x^* + \varepsilon \Rightarrow \text{plim } \hat{b} = \lambda \beta, \quad \lambda = \frac{\text{Var}(x^*)}{\text{Var}(x^*) + \text{Var}(u)}$$

- ▶ $\lambda \in (0, 1]$: the slope is biased towards zero (attenuation); $1/\lambda$ undoes it

- **Misclassified dummy Aigner (1973)**: label $\hat{s} \in \{0, 1\}$ for a true $s^* \in \{0, 1\}$

- ▶ $\pi = P(s^* = 1)$; false alarms $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$; misses $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$
- ▶ Total probability: $p = P(\hat{s} = 1) = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$
- ▶ $u = \hat{s} - s^*$ can be -1 only if $s^* = 1$: correlated with s^* , not classical; A1 derives λ
- ▶ Non-differential error: given s^* , $\hat{s}$ carries no extra information on r

What You Need for Today (7/20): Attenuation in a Simulation



- Left: 400 simulated texts, $\pi = 0.25$, $\alpha_0 = 0.15$, $\alpha_1 = 0.25$, true $\beta = 1$: the slope on the label is about half the slope on the truth
- Right: λ against the error rate; $\lambda = 0$ at 50% errors (a coin toss); a rare class (small π) loses more

What You Need for Today (8/20): Two Classifiers on the Same Texts

- Paired data: for text i , $D_i = 1$ if A is wrong and B right, -1 if A is right and B wrong, 0 otherwise
 - ▶ n_{01} , n_{10} : numbers of texts with $D_i = 1$ and $D_i = -1$ (the discordant texts); texts both get right or both get wrong give 0

- **Accuracy gap** and its variance (derived in A2)

$$\hat{d} = \frac{n_{01} - n_{10}}{n}, \quad \text{Var}(\hat{d}) = \frac{q - d^2}{n}$$

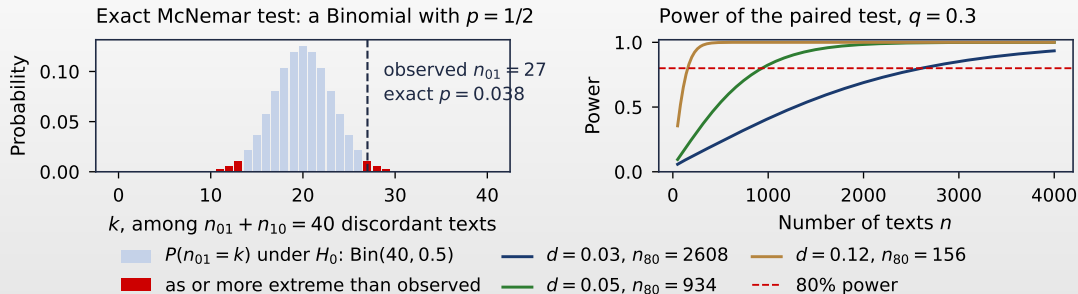
- ▶ $d = \text{acc}_B - \text{acc}_A$; $q = P(D_i \neq 0)$, the share of discordant texts: only they add variance

- **Exact McNemar test** McNemar (1947): under H_0 (equal accuracy) each discordant text favours B with probability $\frac{1}{2}$

$$n_{01} \sim \text{Bin}\left(m, \frac{1}{2}\right), \quad m = n_{01} + n_{10}, \quad p = \min\left\{1, 2P\left(\text{Bin}\left(m, \frac{1}{2}\right) \geq \max(n_{01}, n_{10})\right)\right\}$$

- ▶ Large-sample version: $\chi^2 = (n_{01} - n_{10})^2 / m$, compared with 3.84, the 95% quantile of χ_1^2

What You Need for Today (9/20): The McNemar Test and Its Power

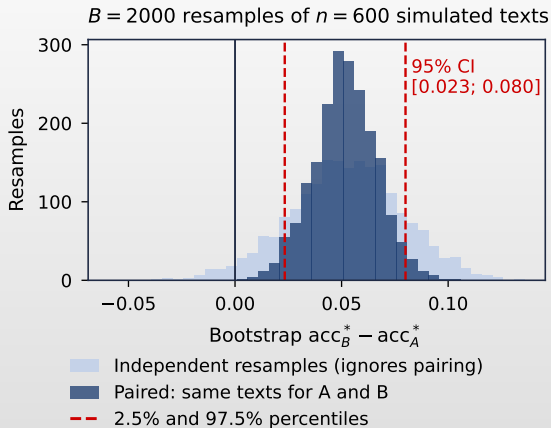


- Power of the two-sided 5% test if the true gap is d , and the n that gives 80% power:

$$\text{power} \approx \Phi\left(\frac{|d|\sqrt{n}}{\sqrt{q - d^2}} - 1.96\right), \quad n_{80} = \frac{(1.96 + 0.84)^2 (q - d^2)}{d^2}$$

- Φ : standard Normal distribution function; 0.84: its 80% quantile; left: red bars = the exact p -value for $m = 40$, $n_{01} = 27$

What You Need for Today (10/20): Bootstrap Confidence Intervals



- ▣ **Percentile bootstrap** for a statistic $\hat{\theta}$ (an accuracy, a gap)
 - ▶ Draw n texts with replacement, B times; recompute $\hat{\theta}_b^*$, $b = 1, \dots, B$
 - ▶ 95% CI = $[q_{2.5\%}, q_{97.5\%}]$, the percentiles of the $\hat{\theta}_b^*$
- ▣ **Paired bootstrap**: the same resampled texts for A and B
 - ▶ Keeps the difficulty each text shares between A and B: a narrower interval for the gap (chart)
- ▣ **Block bootstrap** for days (C2)
 - ▶ Resample blocks of consecutive days, which keeps their serial dependence

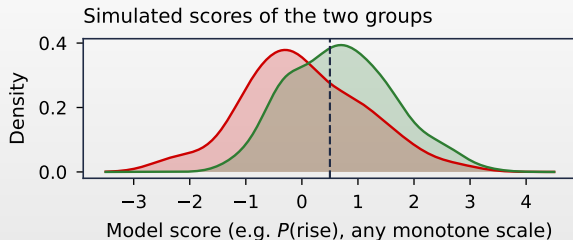
What You Need for Today (11/20): Ranking Quality: AUC

- Task of **B3**: separate “up” months (n_1) from “down” months (n_0) with a score S , e.g. $P(\text{rise})$
- **AUC**: probability that a random up month scores above a random down month

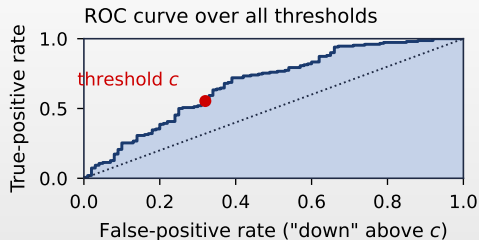
$$\text{AUC} = \frac{U}{n_1 n_0}, \quad U = \sum_{i \in \text{up}} \sum_{j \in \text{down}} \left(1\{S_i > S_j\} + \frac{1}{2} 1\{S_i = S_j\} \right)$$

- ▶ U : Mann–Whitney statistic, the number of correctly ordered pairs; $1\{\cdot\}$: 1 if true, else 0
- ▶ $\text{AUC} = 0.5$: no information; 1: perfect ordering; only the order of the scores matters
- **Permutation test** of $\text{AUC} = 0.5$
 - ▶ Shuffle the up/down labels B times; $p = (1 + \#\{\text{AUC}_b^* \geq \text{AUC}\})/(B + 1)$, $\#$: count
- **Two independent AUCs** (before and after a date), DeLong standard errors [DeLong et al. \(1988\)](#)
 - ▶ $z = (\text{AUC}_1 - \text{AUC}_2) / \sqrt{\text{SE}_1^2 + \text{SE}_2^2} \sim N(0, 1)$ under H_0 ; gap detectable with 80% power at 5%: $2.80 \sqrt{\text{SE}_1^2 + \text{SE}_2^2}$

What You Need for Today (12/20): Scores, Thresholds and the ROC Curve



"down" months --- a threshold c
 "up" months area = AUC = 0.68

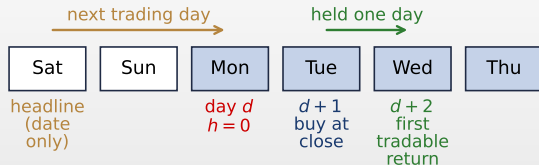


ROC curve
 no information: AUC = 0.5

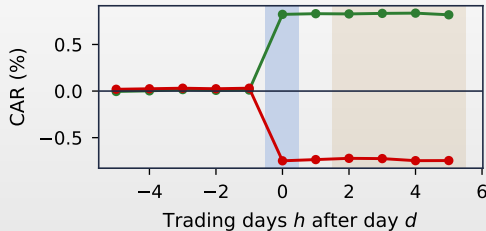
- Left: simulated scores of 150 up and 100 down months; a threshold c calls every month above it "up"
- Right: moving c traces the ROC curve, true-positive against false-positive rate; the area under it is the AUC (here 0.68)

What You Need for Today (13/20): From Headlines to Returns: Timing

From a calendar date to the tradable return



Simulated: all news priced on day d



—●— CAR, positive news —●— CAR, negative news ■ day d : reaction ■ $h \geq 2$: prediction

- Day d : the first trading day on or after the headline date; the hour is unknown, so $d + 1$ may still hold the reaction
- A slope at $h = 0$ measures how headlines *describe* the day; only $h \geq 2$ is a *forecast* (B2)
- Event study (B11): $CAR[h_1, h_2] = \sum_{h=h_1}^{h_2} \bar{r}_{d+h}^e$, the cumulated mean excess return; right: news fully priced on day d

What You Need for Today (14/20): Panel Regression and Clustered Standard Errors

- Panel regression on stock-days (stock i , day d , horizon h)

$$r_{i,d+h}^e = a + b z_{i,d} + \varepsilon_{i,d}$$

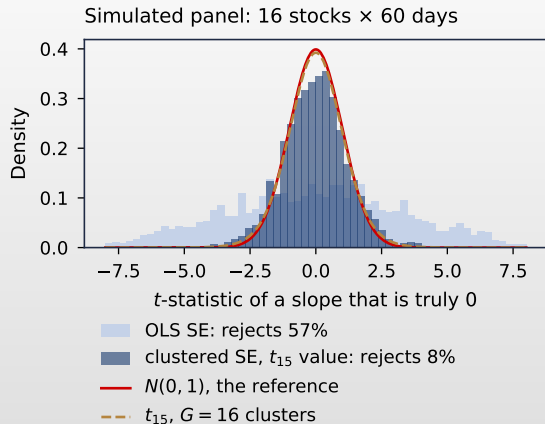
- ▶ r^e : excess return over SPY; $z_{i,d}$: daily score standardised over the panel; b : return per one SD of the score

- Errors are correlated within a day (common news) and within a stock (over time): OLS standard errors are too small
- **Clustered SE Petersen (2009)**: sum the scores within each cluster g (a day or a stock)

$$\widehat{\text{Var}}(\hat{\beta}) = c (X^\top X)^{-1} \left(\sum_{g=1}^G X_g^\top \hat{u}_g \hat{u}_g^\top X_g \right) (X^\top X)^{-1}, \quad c = \frac{G}{G-1} \frac{n-1}{n-2}$$

- ▶ $\hat{\beta} = (\hat{a}, \hat{b})$; X : the n rows $(1, z_{i,d})$; $X_g, \hat{u}_g$: rows and OLS residuals of cluster g ; c : small-sample factor
- ▶ Few clusters ($G = 16$ stocks): compare $t = \hat{b}/\text{SE}$ with t_{G-1} (2.13), not with 1.96, or use a wild cluster bootstrap Cameron, Gelbach & Miller (2008)

What You Need for Today (15/20): Why Clustering Matters



- Simulation: 16 stocks $\times$ 60 days, true slope 0; both the score and the error persist within a stock
- OLS SE treat the 960 rows as independent
 - The t -statistic is far too dispersed: 57% false rejections at 5%
- Clustered SE with the t_{15} critical value
 - About 8% false rejections: close to, but still above, 5%
 - With so few clusters, a wild cluster bootstrap is safer (B9)

What You Need for Today (16/20): A Sentiment Strategy, Its t and Its Costs

- Strategy (A3): at the close of $d + 1$, a position of sign $s_{i,d}$ in each stock with news, hedged with SPY, closed a day later; R_t : its return on day t
- Newey–West t Newey & West (1987): the t of a mean when days are serially dependent

$$t_{NW} = \frac{\hat{\mu}}{\sqrt{\hat{\Omega}/T}}, \quad \hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) \hat{\gamma}_l$$

► $\hat{\mu} = \bar{R}$; $\hat{\gamma}_l$: sample autocovariance of R_t at lag l ; L : number of lags; T : number of days

- Costs: c per trade, four trades on each day with positions, a share a of such days

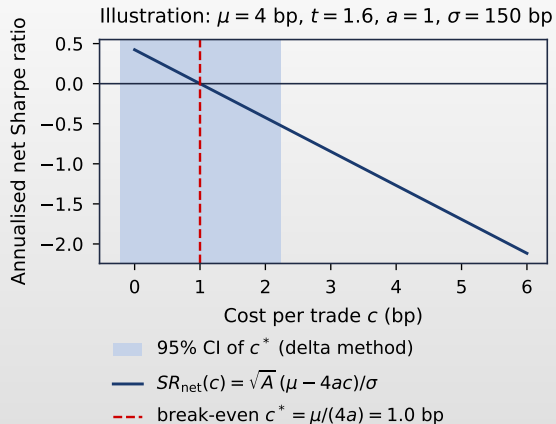
$$\mu_{\text{net}} = \mu - 4ac, \quad SR_{\text{net}} = \frac{\sqrt{A}(\mu - 4ac)}{\sigma}, \quad c^* = \frac{\mu}{4a}$$

► $A = 252$ trading days a year; σ : SD of daily returns; SR : Sharpe ratio; c^* : break-even cost

- Delta method: the SE of a function of an estimate

► $SE(g(\hat{\mu})) \approx |g'(\hat{\mu})| SE(\hat{\mu})$; with $SE(\hat{\mu}) = \hat{\mu}/t_{NW}$ and a fixed: $SE(c^*) = SE(\hat{\mu})/(4a)$

What You Need for Today (17/20): Break-Even Cost: an Illustration

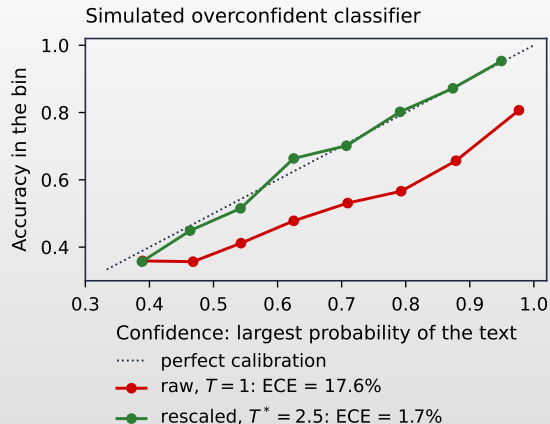


- Illustrative inputs, not those of A3: $\mu = 4$ bp a day, $t_{NW} = 1.6$, $a = 1$, $\sigma = 150$ bp
- The net Sharpe ratio falls linearly in c and crosses 0 at $c^* = 4/4 = 1$ bp
- $SE(c^*) = (4/1.6)/4 = 0.63$ bp
 - ▶ 95% CI of c^* : $1 \pm 1.96 \times 0.63 = [-0.23; 2.23]$ bp
 - ▶ A realistic cost of 5 bp per trade lies far to the right of the interval

What You Need for Today (18/20): Look-Ahead Bias and Many Tries

- **Look-ahead bias** [Glasserman & Lin \(2023\)](#): an LLM trained on text up to its release may have read what followed a headline
 - ▶ A backtest on news before the release mixes forecasting with remembering
 - ▶ Memory test ([B3](#)): ask the model about past months without any data; an AUC above 0.5 means it remembers
 - ▶ Clean test: only texts published after the weights were released ([B3](#), [C1](#))
- **Many tries**: several scores, horizons or prompts, then the best one reported
 - ▶ With 20 independent tests at 5%, the chance of at least one false rejection is $1 - 0.95^{20} = 0.64$
 - ▶ Report every try, or correct the p -values ([B7](#), [B9](#), [B10](#))
- More in [Chapter 15](#): embeddings, calibration, the scores as estimates and the ChatGPT case study

What You Need for Today (19/20): For the Extension Packs: Calibration



- Calibration (A7) [Guo et al. \(2017\)](#): among texts given probability 0.8, about 80% should be right
- Loss used to choose T
 - ▶ $\text{NLL}(T) = -\sum_i \ln p_{iy_i}(T)$, y_i : true label of text i
- Brier score: mean squared error of the probabilities
 - ▶ $\frac{1}{n} \sum_i \sum_k (p_{ik} - 1\{y_i = k\})^2$
- ECE: $\sum_b \frac{n_b}{n} |\text{acc}_b - \text{conf}_b|$
 - ▶ Bins b of the largest probability; conf_b : its mean, acc_b : accuracy, n_b : texts in the bin
- Chart: an overconfident model lies below the diagonal; $T^* > 1$ moves it onto it

What You Need for Today (20/20): For the Extension Packs: Inference Tools

- **Holm correction** for M tests: sort $p_{(1)} \leq \dots \leq p_{(M)}$
 - ▶ $\tilde{p}_{(j)} = \max_{i \leq j} \min\{1, (M - i + 1) p_{(i)}\}$; reject where $\tilde{p} < 0.05$: at most 5% chance of any false rejection
- **Wild cluster bootstrap** [Cameron, Gelbach & Miller \(2008\)](#) (B9)
 - ▶ $y_i^* = \bar{y} + w_{g(i)}(y_i - \bar{y})$, $w_g = \pm 1$ with probability $\frac{1}{2}$ per cluster; $p = P^*(|t^*| \geq |t|)$
 - ▶ Two-way clustering: $\hat{V} = \hat{V}_{\text{day}} + \hat{V}_{\text{stock}} - \hat{V}_{\text{day} \cap \text{stock}}$
- **Prediction-powered inference** [Angelopoulos et al. \(2023\)](#) (B8): a model mean corrected by its bias on labelled texts
 - ▶ $\hat{\theta} = \frac{1}{N-n} \sum_{\text{unlab.}} f - \frac{1}{n} \sum_{\text{lab.}} (f - Y)$; f : model score, Y : human label
- **Fleiss' κ** [Fleiss \(1971\)](#) (A5): agreement among raters beyond chance
 - ▶ $\kappa = (\bar{P} - \bar{P}_e) / (1 - \bar{P}_e)$: $\bar{P}$ observed, $\bar{P}_e$ chance agreement; 0 = chance, 1 = perfect

A1 [Solved]: Attenuation from Misclassification — Task

- **Question:** a regression uses the FinBERT label "negative" instead of the truth; by how much is the slope shrunk?
- **Data**
 - ▶ Twitter Financial News, validation set: $N = 2\,388$ headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
 - ▶ Model $r = a + \beta s^* + \varepsilon$, $s^* = 1$ if the headline is truly negative; we observe only $\hat{s} = 1$ when FinBERT predicts negative
 - ▶ Non-differential error: given s^* , $\hat{s}$ says nothing about r , so $\text{Cov}(\varepsilon, \hat{s}) = 0$
 - ▶ π , p , false alarms α_0 and misses α_1 : the rates defined in [What You Need \(6/20\)](#)
- **Tasks**
 - (a) Show that $\text{plim } \hat{b} = \beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$.
 - (b) Derive $\lambda = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1) / [p(1 - p)]$.
 - (c) Compute π , p , α_0 , α_1 from the matrix, then λ to three decimals.
 - (d) Compute $\text{Cov}(u, s^*)$ for $u = \hat{s} - s^*$: is the error classical?
- **Report:** λ , $1/\lambda$ and the sign of $\text{Cov}(u, s^*)$
- **Interpretation:** what share of a true effect survives in a slope on the FinBERT dummy?

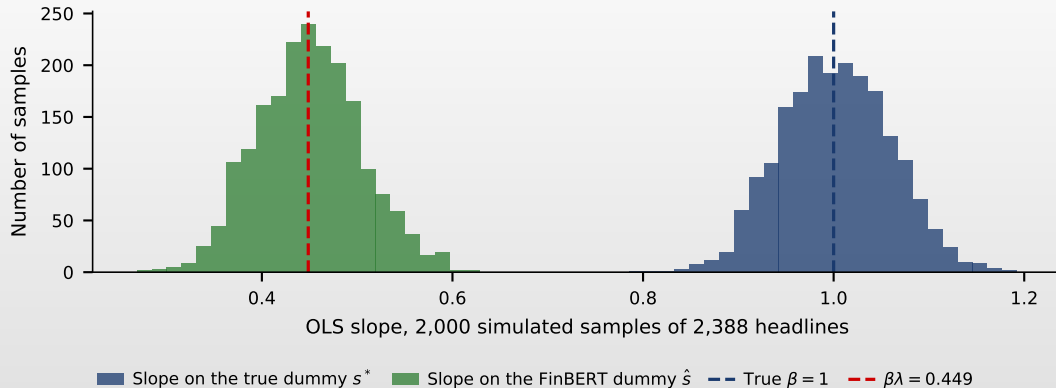
FinBERT confusion matrix, Twitter

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

A1 [Solved]: Solution (1/3)

- ▣ (a) For one regressor $\text{plim } \hat{b} = \text{Cov}(r, \hat{s}) / \text{Var}(\hat{s})$ and $\text{Cov}(r, \hat{s}) = \beta \text{Cov}(s^*, \hat{s}) + \text{Cov}(\varepsilon, \hat{s})$; the last term is 0 by the non-differential assumption
- ▣ (b) Both variables are binary: $\text{Cov}(s^*, \hat{s}) = P(s^* = 1, \hat{s} = 1) - \pi p = \pi(1 - \alpha_1) - \pi p$
- ▣ Total probability: $p = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$; substituting, $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$
- ▣ $\hat{s}$ is Bernoulli, so $\text{Var}(\hat{s}) = p(1 - p)$; the ratio of the two is λ

A1 [Solved]: Solution (2/3)



A1 [Solved]: Solution (3/3)

- (c) $\pi = 347/2\,388 = 0.1453$, $p = 535/2\,388 = 0.2240$, $\alpha_1 = 1 - 264/347 = 0.2392$, $\alpha_0 = 271/2\,041 = 0.1328$
- $\lambda = \frac{0.1242 \times 0.6280}{0.2240 \times (1 - 0.2240)} = 0.449$, so $\beta = 2.23 \hat{b}$
- (d) $\text{Cov}(u, s^*) = 0.0780 - 0.1242 = -0.0462 < 0$: the error is negatively correlated with the truth, not classical [Aigner \(1973\)](#)
- Chart: 2 000 simulated samples with these rates and $\beta = 1$; the slope on $\hat{s}$ averages 0.450 (SD 0.054), as λ predicts
- Interpretation: a slope on the FinBERT dummy keeps only about 45% of the true effect

 MFM_ch15_sem_partA

A2 [Solved]: Paired Accuracy Difference: Variance, Exact Test, Power — Task

▣ Questions

- ▶ how precise is the accuracy gap between two classifiers scored on the same headlines?
- ▶ how many headlines does it need?

▣ Inputs

- ▶ Twitter validation set, $n = 2\,388$ independent headlines; A = LM dictionary, B = FinBERT
- ▶ Discordant counts: $n_{01} = 539$ (A wrong, B right), $n_{10} = 252$ (A right, B wrong)
- ▶ $D_i = 1$ if A wrong and B right, -1 if A right and B wrong, 0 otherwise; $d = ED_i = \text{acc}_B - \text{acc}_A$, $q = P(D_i \neq 0)$

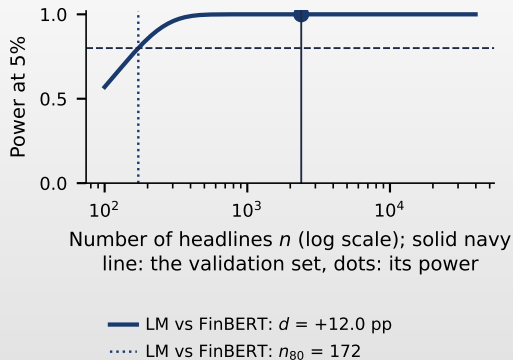
▣ Tasks

- Show $\hat{d} = (n_{01} - n_{10})/n$ and derive $\text{Var}(\hat{d}) = (q - d^2)/n$.
- Compute $\hat{d}$, SE, the 95% CI and $z = \hat{d}/\text{SE}$.
- Run the exact McNemar test at 5% and state the decision.
- If the true d is $\hat{d}$, compute the power and n_{80} , the n for 80% power, with the formulas of [What You Need \(9/20\)](#).

▣ **Report:** $\hat{d}$, SE, CI, z , exact p , the decision, the power and n_{80}

▣ **Interpretation:** which headlines determine the variance of $\hat{d}$?

A2 [Solved]: Solution



- (a) $\hat{d} = \frac{1}{n} \sum_i D_i = (n_{01} - n_{10})/n$; $D_i^2 = 1$ exactly when one is right and the other wrong, so $ED_i^2 = q$, $\text{Var}(D_i) = q - d^2$ and $\text{Var}(\hat{d}) = (q - d^2)/n$
- (b) $\hat{d} = (539 - 252)/2388 = 0.1202$,
 $\hat{q} = 791/2388 = 0.3312$, $\text{SE} = 1.15$ pp, $\text{CI} [9.76; 14.28]$ pp, $z = 10.43$
- (c) Under H_0 we expect 395.5 on each side; exact $p < 0.001$: reject equal accuracy
- (d) Power $\Phi(+8.47) \approx 100\%$;
 $n_{80} = 7.84 \times (0.3312 - 0.01444)/0.01444 \approx 170$ headlines (chart)
- Interpretation: only the discordant headlines (q) enter the variance; a 12-point gap is detectable with a few hundred headlines

A3 [Solved]: Break-Even Cost with a Standard Error — Task

▣ Questions

- ▶ at what cost per trade does the FinBERT strategy stop earning?
- ▶ how precisely do we know that cost?

▣ Inputs

- ▶ Sentiment strategy ([What You Need \(16/20\)](#); studied in [Chapter 15](#)): at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
- ▶ FinBERT signal: mean gross return over all days $\hat{\mu} = 3.27$ bp, $t_{NW} = 1.27$ ($L = 8$ lags), share of days with positions $a = 0.962$, gross Sharpe ratio $\sqrt{A} \hat{\mu} / \sigma = 0.36$, σ the standard deviation of daily returns, $A = 252$ trading days a year

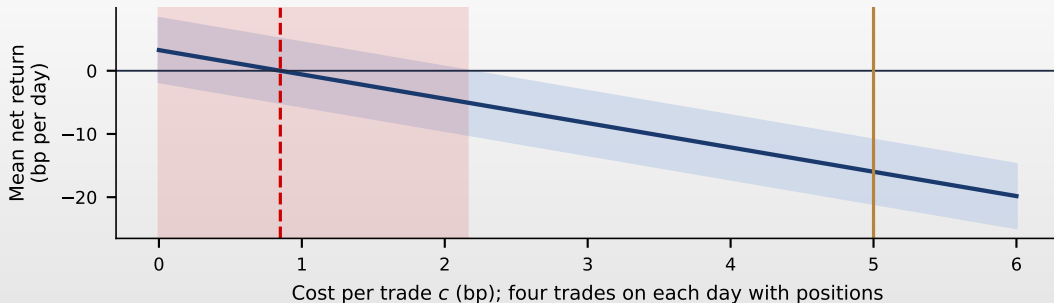
▣ Tasks

- Write the mean net daily return and the net Sharpe ratio as functions of the cost c per trade.
- Derive the break-even cost c^* at which the mean net return is zero.
- Compute $\text{SE}(\hat{\mu}) = \hat{\mu} / t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI of c^* .
- Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp.

▣ **Report:** c^* , its SE and CI, the net Sharpe ratio

▣ **Interpretation:** is a realistic cost of 5 bp per trade inside the CI of c^* ?

A3 [Solved]: Solution (1/2)



±1.96 SE of the mean

--- Break-even cost $c^* = 0.85$ bp

— Realistic cost, 5 bp per trade

— FinBERT strategy: mean net return per day

95% CI of c^*

A3 [Solved]: Solution (2/2)

- (a) On a share a of days we pay $4c$: $\mu_{\text{net}} = \mu - 4ac$, $SR_{\text{net}} = \sqrt{A}(\mu - 4ac)/\sigma$, with σ barely changed by costs
- (b) $c^* = \mu/(4a) = 3.27/3.85 = 0.85$ bp per trade
- (c) $SE(\hat{\mu}) = 3.27/1.27 = 2.57$ bp; with a fixed by the news calendar, $SE(c^*) = 2.57/3.85 = 0.67$ bp, CI $[-0.46; 2.16]$ bp
- (d) $\sigma = \sqrt{252} \times 3.27/0.36 = 143$ bp
 - ▶ $SR_{\text{net}} = \sqrt{252}(3.27 - 3.85 \times 5)/143 = -1.77$
- Interpretation: 5 bp lies far above the CI of c^* ; with daily turnover the strategy cannot survive realistic costs

B1 [Solved]: Is FinBERT Better than LM? — Task

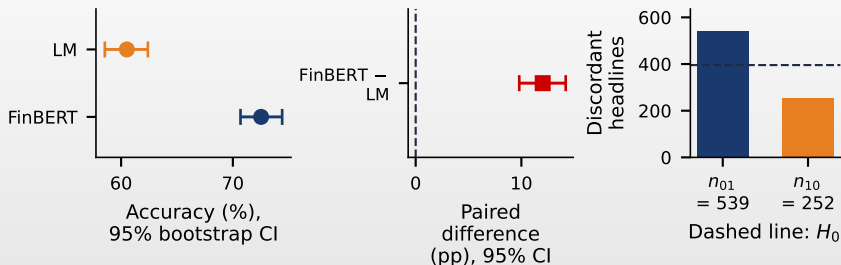
- **Question:** is the fine-tuned FinBERT more accurate than the Loughran–McDonald dictionary on headlines it was not trained on?
- **Data**
 - ▶ Twitter validation set of **A1**: $N = 2\,388$ headlines, human label = truth
 - ▶ A = LM dictionary (label = sign of the tone), B = FinBERT (label = largest probability); notebook function `b_pairs`
- **Tasks**
 - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2 000 times (seed 0).
 - (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers.
 - (c) Count n_{01} , n_{10} ; run the exact McNemar test and $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ_1^2 (3.84).
- **Report:** a table of the two accuracies, their CIs and the difference; n_{01} , n_{10} , both p -values and the decision
- **Interpretation**
 - ▶ why is the paired test more powerful than comparing the two intervals?
 - ▶ does the gap matter for trading?

B1 [Solved]: Solution (1/2)

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

- Step 1: labels of both classifiers on the 2 388 headlines; step 2: 2 000 bootstrap resamples of the headline indices; step 3: both accuracies and their difference in each resample
- (a)–(b) The two intervals do not overlap and the paired CI of the difference excludes 0; it agrees with the analytic CI of A2

B1 [Solved]: Solution (2/2)



- (c) $n_{01} = 539$, $n_{10} = 252$, 395.5 expected under H_0 ; exact $p < 0.001$; $\chi^2 = 104.1 > 3.84$: reject equal accuracy
- Pairing removes the variation in headline difficulty: headlines both get right, or both get wrong, carry no information
- Interpretation: FinBERT is clearly better than the dictionary outside its training data; a 12-point gain in labelling says nothing yet about returns (B2)

B2 [Solved]: Reaction or Prediction? — Task

▣ Questions

- ▶ does the daily headline score explain the return of the news day?
- ▶ does it forecast the first return one can still trade?
- ▶ how wide are the honest intervals?

▣ Data

- ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023; $z_{i,d}$: daily net score, standardised over the panel; r^e : excess return over SPY (%), from the adjusted daily closes of the course data
- ▶ Scores: FinBERT, Qwen2.5-7B, LM

▣ Tasks

- Estimate $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$ by OLS for the three scores and $h = 0, 1, 2$; report b in bp per one SD of the score.
- Compute three SEs: OLS, clustered by day ($G = 3\,434$), clustered by stock ($G = 16$), with the small-sample factor c of [What You Need \(14/20\)](#).
- Compare each t with 1.96, and the stock-clustered t with the t_{15} critical value 2.13.

- ▣ **Report:** a chart of the $h = 0$ slopes with the three intervals, and a table of the regressions

▣ Interpretation

- ▶ why do $h = 0$ and $h = 2$ differ?
- ▶ can a clustered SE with 16 clusters be trusted?

B2 [Solved]: Solution (1/3)



B2 [Solved]: Solution (2/3)

- Same-day slopes with the 95% intervals of the three standard errors: the stock-clustered interval is the widest
- Clustering by day is close to OLS: with only 5.3 stocks per day there is little common noise
- Clustering by stock multiplies the FinBERT SE at $h = 0$ by 3.0: the slope differs across stocks and each stock's errors are correlated over time

 MFM_ch15_sem_partB

B2 [Solved]: Solution (3/3)

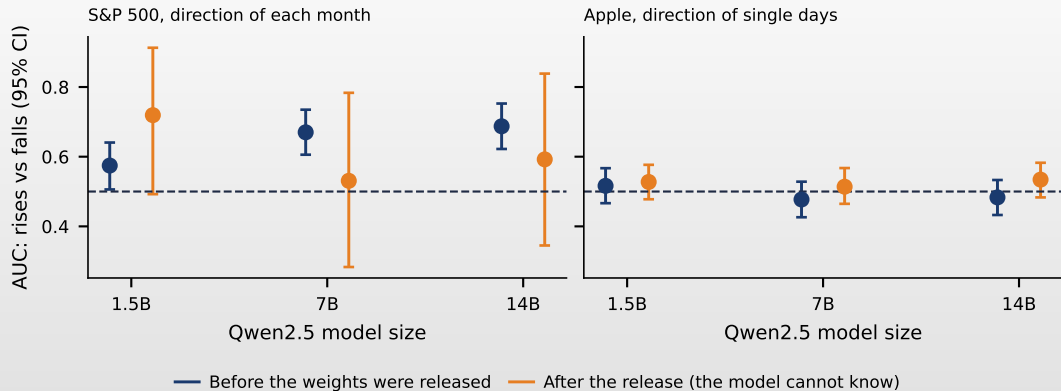
Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

- $h = 0$: large under every SE (t by stock above 4.7): headlines describe the day's move, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$ by day); Qwen $t = +2.41$ by stock, but $p = 0.061$ by wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#) and 0.37 after Holm over nine regressions (extension [B9](#))
- Interpretation: sentiment explains the news day, not the tradable days; with 16 stocks, inference needs few-cluster methods

B3 [Solved]: Does the Model Remember the Market? — Task

- **Question:** asked without any data, can Qwen2.5 tell the months in which the S&P 500 rose from those in which it fell?
- **Data**
 - ▶ S&P 500 monthly closes (GSPC.INDX), January 2000 to August 2026; a month is "up" if its return is positive
 - ▶ Qwen2.5 1.5B, 7B, 14B (weights published 19 September 2024) asked "Did the S&P 500 index rise or fall during ⟨month year⟩?"; score $P(\text{rise})$
 - ▶ Before: 296 months to August 2024 (185 up); after: 23 months from October 2024 (13 up)
- **Tasks**
 - (a) Compute the $\text{AUC} = U/(n_{\text{up}}n_{\text{down}})$ (What You Need (11/20)) per model and period, with a 2000-draw bootstrap CI over months.
 - (b) Permutation test for 14B before the release: shuffle the up/down labels 2000 times.
 - (c) Test $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$ with DeLong variances DeLong et al. (1988) (What You Need (11/20)) and find the smallest drop detectable with 80% power.
- **Report:** the AUC table with CIs, the permutation p , the DeLong z and p , the detectable drop
- **Interpretation:** what does an AUC above 0.5 mean when the model sees no data?

B3 [Solved]: Solution (1/3)

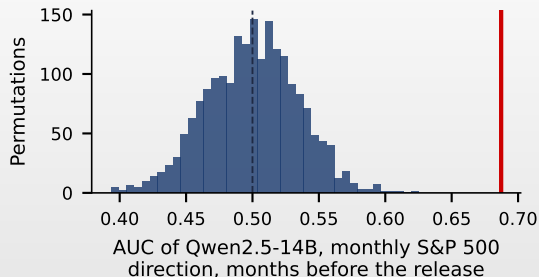


B3 [Solved]: Solution (2/3)

Qwen2.5	AUC before	CI 95%	AUC after	CI 95%
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]

- (a) 7B and 14B rank up months above down months far above chance before the release; 1.5B barely; after the release all intervals contain 0.5

B3 [Solved]: Solution (3/3)



- AUC with shuffled up/down labels (2,000 permutations)
- Observed AUC = 0.69

- (b) Permutation p for 14B before the release < 0.001 : no shuffled AUC reaches the observed one (chart)
- (c) DeLong: SE before 0.032, after 0.126; $z = 0.73$, $p = 0.47$: do not reject equality
 - ▶ Detectable drop $2.80 \sqrt{SE_{\text{before}}^2 + SE_0^2} = 0.36$
 - ▶ About 101 post-release months would be needed
- Interpretation: without data, the discrimination comes from the training text
 - ▶ A backtest on pre-release news mixes forecasting with remembering [Glasserman & Lin \(2023\)](#)

C1 [Proposed]: Critique an AI Answer — the Answer

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Code of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of [What You Need \(1/20\)](#) and [What You Need \(2/20\)](#))

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C1 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** B1, B2 and B3 [Solved]
- ▣ **Conclusion of the AI assistant**
 - ▶ “Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well.”
 - ▶ “Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day’s headlines forecast the stock’s excess return.”
 - ▶ “Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample.”
- ▣ **Question:** the code runs and the numbers are right, yet the answer contains **three** errors typical of sentiment backtests; which?
- ▣ **Tasks**
 - ▶ (a) Find each error: name the line of code or the sentence and explain why it is wrong.
 - ▶ (b) Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test.
 - ▶ (c) Write the corrected code or statement, with the numbers it produces.
 - ▶ (d) Log each error in [AI_ERRORS.md](#) with the fields: what the AI answered, what was wrong, how you found out, the correct version.

B4 [Proposed]: Is a 7-Billion-Parameter LLM Better than FinBERT? — Task

□ **Model:** B1 [Solved]

□ **Question:** does a general open LLM, used zero-shot, beat the fine-tuned FinBERT on headlines neither was trained on?

□ **Data**

- ▶ Twitter validation set of A1: $N = 2\,388$ headlines, human label = truth
- ▶ $A = \text{FinBERT}$, $B = \text{Qwen2.5-7B}$ zero-shot with prompt P1; neither used any Twitter label

□ **Tasks**

- Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2000 times (seed 0).
- Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers.
- Count n_{01} , n_{10} ; run the exact McNemar test and $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ_1^2 (3.84).

□ **Report:** the same table, tests and chart as in B1

□ **Interpretation**

- ▶ do the two intervals overlap?
- ▶ does that settle the question?
- ▶ what would a much larger, slower model still bring?

Team Project Step

- Choose a text source not covered in class (BVB announcements, Romanian financial news, central-bank statements, crypto posts)
- Label at least 200 texts with two annotators and report the agreement; evaluate only on a held-out part
- Log AI use in [AI_USE.md](#) and caught errors in [AI_ERRORS.md](#); extension packs are optional; C2 and C3 are project starts

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Measurement error	A4, A5	A1
2. Comparing classifiers	A6, B5 [Solved], B6, B7	A2, B1
3. LLM probabilities	A7 [Solved]	What You Need (2/20) and What You Need (19/20), lecture calibration chart
4. Valid inference	B8 [Solved], B9, B10	B2, B3
5. Signals and costs	A8, B11, C2	A3, B2
6. Replication lab	C3	C2, the lecture case study

- Each exercise has its own notebook section; do them in order within a pack
- Useful for the team project: C2 and C3 are possible project starts

A4 [Proposed]: Attenuation for the LM Dictionary — Task

- **Model:** A1 [Solved]
- **Question:** is the weaker same-day LM slope of Lecture 15 (15.5 bp against 39.2 bp for FinBERT) a weaker market reaction, or a worse measurement?
- **Data**
 - ▶ Twitter validation set of A1: $N = 2\,388$ headlines, human label = truth
 - ▶ Same model and assumption as A1; now $\hat{s} = 1$ when the LM tone of the headline is negative
- **Tasks**
 - (a) Compute π , p , α_0 and α_1 from the LM matrix.
 - (b) Compute λ and $1/\lambda$ with the formula of A1.
 - (c) Compute $\text{Cov}(u, s^*)$: is the LM error classical?
 - (d) Compare λ with that of FinBERT (A1) and compute the ratio of the two.
- **Report:** the four rates, λ , $1/\lambda$, the ratio
- **Interpretation:** which classifier attenuates the slope more?

LM confusion matrix, Twitter

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

A5 [Proposed]: Fleiss' κ and One Dawid–Skene Step — Task

▣ **Model:** A1 [Solved]

▣ **Questions**

- ▶ how much do raters agree beyond chance?
- ▶ what is the most likely true label of a disputed sentence?

▣ **Inputs**

- ▶ Illustrative votes (made up for the exercise): 6 sentences, $m = 5$ raters each
- ▶ $P_i = \sum_j n_{ij}(n_{ij} - 1)/[m(m - 1)]$, $\bar{P} = \frac{1}{N} \sum_i P_i$, p_j = share of votes for j ,
 $\bar{P}_e = \sum_j p_j^2$; κ Fleiss (1971): What You Need (20/20)

▣ **Tasks**

- (a) Compute every P_i and $\bar{P}$.
- (b) Compute p_j , $\bar{P}_e$ and κ .
- (c) E-step of Dawid & Skene (1979): each rater gives the true class with probability 0.8 and each other class with 0.1, uniform prior; compute $P(\text{true} = j \mid \text{votes}) \propto \prod_i \pi_{ji}^{n_{ij}}$ for sentences 4 and 5.
- (d) Compare the posteriors with the vote shares and describe the M-step.

▣ **Report:** κ and the two posteriors

▣ **Interpretation:** how would you treat the disputed sentences in a test set?

Sentence	neg	neu	pos
1	5	0	0
2	0	5	0
3	0	1	4
4	1	3	1
5	2	3	0
6	0	2	3

A6 [Proposed]: Is "No Difference" a Power Problem? — Task

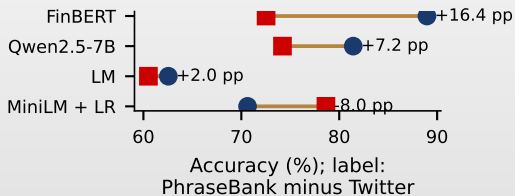
- **Model:** A2 [Solved]
- **Question:** FinBERT and Qwen2.5-7B look tied: is that evidence of equal accuracy, or of a weak test?
- **Inputs**
 - ▶ Twitter validation set, $n = 2\,388$; $A = \text{FinBERT}$, $B = \text{Qwen2.5-7B (prompt P1)}$; $n_{01} = 395$, $n_{10} = 355$
- **Tasks**
 - (a) Compute $\hat{d}$, $\hat{q}$, SE, the 95% CI and z with the formulas of A2.
 - (b) Run the exact McNemar test at 5%.
 - (c) Compute the power at $n = 2\,388$ if the true d is $\hat{d}$, and n_{80} .
 - (d) Use the CI to state which differences the data rule out.
- **Report:** the numbers of (a)–(d)
- **Interpretation:** may you conclude that the two models are equally accurate?

B5 [Solved]: Training-Data Leakage — Task

- ▣ **Question:** how much of FinBERT's accuracy on Financial PhraseBank comes from having been trained on it?
- ▣ **Data**
 - ▶ Financial PhraseBank [Malo et al. \(2014\)](#): 4 846 sentences (2 264 with full agreement); FinBERT was fine-tuned on a split of them (3 101 of 4 845 in [Araci \(2019\)](#))
 - ▶ Twitter validation set, 2 388 headlines unseen by FinBERT; classifiers: FinBERT, Qwen2.5-7B, LM, MiniLM + logistic regression trained on Twitter [Wang et al. \(2020\)](#)
- ▣ **Tasks**
 - (a) Compute accuracy (2 000-draw bootstrap CI) and macro-F1 on PhraseBank with full agreement, all of PhraseBank and Twitter.
 - (b) Compute the drop from all of PhraseBank to Twitter for each classifier.
 - (c) Compute the difference in differences FinBERT minus Qwen2.5-7B with a bootstrap CI (each data set resampled independently).
- ▣ **Report:** a table of accuracies and drops, the difference in differences with its CI
- ▣ **Interpretation:** how do you separate "harder data" from "memory of the training set"?

B5 [Solved]: Solution

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	-8.0	69.9



● Financial PhraseBank (all) ■ Twitter validation set

- (c) $16.4 - 7.2 = 9.2$ pp from unrounded accuracies, bootstrap CI [6.7; 11.9]: excludes 0
- Qwen2.5-7B was not tuned on PhraseBank labels, so its drop measures the harder data
 - ▶ If both would lose the same without overlap (an untested assumption), FinBERT's extra 9.2 points come from overlap
- Interpretation: MiniLM moves the other way because it is trained on Twitter: every model is best at home

B6 [Proposed]: How Many Labels Do You Need? — Task

▣ **Model:** B5 [Solved]

▣ **Question:** how many labelled headlines does a simple supervised model need to beat FinBERT?

▣ **Data**

- ▶ Twitter Financial News: training set of 9 543 headlines, validation set of 2 388; benchmark FinBERT, 72.5%, no Twitter label used
- ▶ Logistic regression ($C = 4$) on 384-dimensional MiniLM embeddings; logistic regression ($C = 8$) on TF-IDF features of words and word pairs

▣ **Tasks**

- For $n \in \{100, 200, 400, \dots, 6\,400, 9\,543\}$ draw 10 random training subsets, fit both models and average the validation accuracy.
- Find the smallest n at which each model beats FinBERT on average.
- Report the SD of the embedding accuracy over the 10 repetitions at $n = 100$.

▣ **Report:** the learning curve and its table

▣ **Interpretation:** why do embeddings start higher but grow more slowly?

B7 [Proposed]: Prompt Sensitivity — Task

- ▣ **Model:** B1 [Solved]
- ▣ **Question:** does the answer of the same LLM change when the same question is worded differently?
- ▣ **Data**
 - ▶ Twitter validation set of A1: $N = 2\,388$ headlines, human label = truth
 - ▶ Qwen2.5-7B and 14B with three prompts: P1 "classify the sentiment . . . for investors", P2 "good, bad or neutral for the stock price", P3 "what is the sentiment of this text"
- ▣ **Tasks**
 - (a) Compute accuracy with a 2 000-draw bootstrap CI and the share of "neutral" answers for each prompt and model.
 - (b) Run the exact McNemar test of P1 against P3 for each model.
 - (c) Compute the share of headlines with the same label under all three prompts.
- ▣ **Report:** a table and a chart of the three prompts
- ▣ **Interpretation:** if you picked the best prompt on this set, what would you be doing?

A7 [Solved]: From Label Probabilities to a Score, and the Optimal Temperature — Task

▣ Questions

- ▶ how does the temperature T change an LLM's probabilities, its score and its calibration?
- ▶ which T fits the data?

▣ Inputs

- ▶ One Twitter headline labelled **neutral**; Qwen2.5-7B (P1) gives $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (exactly 1.0×10^{-12})
- ▶ A random half of the validation set ($n = 1\,194$, seed 15): logits $z_{ik} = \ln p_{ik}$, rescaled $p_{ik}(T)$ as in [What You Need \(2/20\)](#), loss $\text{NLL}(T)$ [Guo et al. \(2017\)](#) of [What You Need \(19/20\)](#)

▣ Tasks

- Give the label, the net score s and the logit gap $\ln(p_{\text{neu}}/p_{\text{neg}})$ of the headline, then its probabilities and s at $T = 2$.
- Explain why labels and accuracy do not depend on T , while the Brier score does.
- Derive $\text{NLL}'(T)$, evaluate the first-order condition at $T = 1$ and solve for T^* (Brent on $[0.5, 50]$).
- On the other half, compute ECE and Brier before and after scaling.

▣ Report: the label, s , $s(T)$, T^* , ECE and Brier before and after

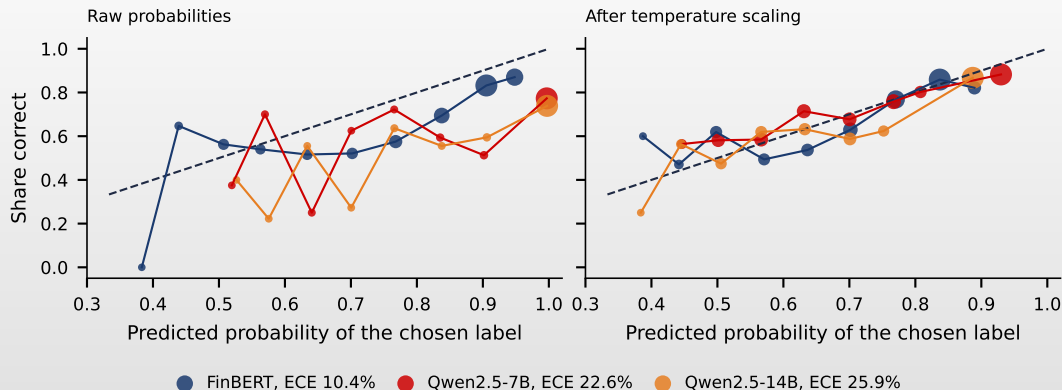
▣ Interpretation

- ▶ is the LLM over- or under-confident?
- ▶ what happens to a rule with a threshold on s ?

A7 [Solved]: Solution (1/3)

- ▣ (a) Largest probability p_{neg} : label negative, wrong; $s = 0.000 - 0.608 = -0.608$; logit gap $\ln(0.392/0.608) = -0.437$
- ▣ At $T = 2$, $p_k(T) \propto \sqrt{p_k}$: $\sqrt{p} = (0.780, 0.626, 1.0 \times 10^{-6})$, sum 1.406, $p(T) = (0.554, 0.446, 0.000)$, $s(T) = -0.554$: same label, score pulled towards 0
- ▣ (b) $z \mapsto z/T$ is increasing for $T > 0$: the order of the logits, hence $\arg \max$ and accuracy, do not change; the probabilities, and with them Brier and log loss, do

A7 [Solved]: Solution (2/3)



A7 [Solved]: Solution (3/3)

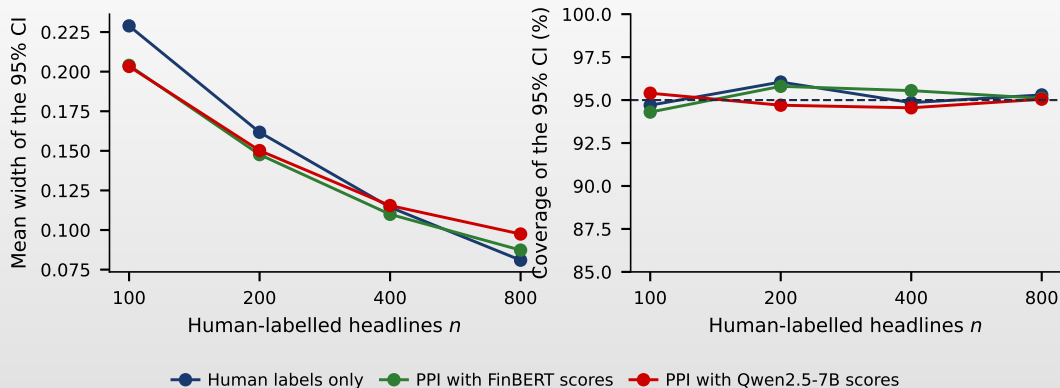
- (c) $\ln p_{iy}(T) = z_{iy}/T - \ln \sum_k e^{z_{ik}/T}$ gives $\text{NLL}'(T) = \frac{1}{T^2} \sum_i (z_{iy_i} - \sum_k p_{ik}(T) z_{ik})$
- At $T = 1$ the average bracket is $-2.47 < 0$, so NLL falls as T grows; the root is $T^* = 8.43$
- (d) ECE falls from 22.6% to 4.3%, Brier from 0.468 to 0.366 (chart: raw and rescaled reliability)
- Interpretation: raw LLM probabilities are heavily overconfident; a threshold on s selects different headlines after rescaling

 MFM_ch15_valid_inference

B8 [Solved]: Prediction-Powered Inference with a Few Human Labels — Task

- **Question:** can a few hundred human labels plus a biased model give a valid and narrower interval for the mean tone of the headlines?
- **Data**
 - ▶ Target θ = mean human net tone $(-1, 0, 1)$ of the 2 388 Twitter validation headlines, which act as the population; f = net score of FinBERT or of Qwen2.5-7B, fixed
 - ▶ Design: n labelled headlines (with Y) and $N - n$ unlabelled ones (only f), drawn independently with replacement
- **Tasks**
 - (a) For one draw with $n = 200$, compute the labels-only estimate $\bar{Y}_n \pm 1.96 s_Y / \sqrt{n}$ ($\bar{Y}_n$, s_Y : mean and standard deviation of the n labels).
 - (b) Compute PPI [Angelopoulos et al. \(2023\)](#): $\hat{\theta}$ of [What You Need \(20/20\)](#), $SE = \sqrt{\hat{V}(f)/(N - n) + \hat{V}(f - Y)/n}$, for both models.
 - (c) For $n = 100, 200, 400, 800$, repeat 2 000 times and record the coverage of θ and the mean CI width.
- **Report:** the one-draw estimates and a coverage and width chart
- **Interpretation:** why is the plain model mean $\frac{1}{N} \sum f$ not an estimate of θ ?

B8 [Solved]: Solution (1/2)



B8 [Solved]: Solution (2/2)

- True $\theta = 0.054$; plain model means FinBERT -0.014 (wrong sign), Qwen $+0.101$: the bias does not shrink as N grows
- (a) Labels only: $0.010 \pm 1.96 \times 0.042$; (b) PPI-FinBERT: rectifier -0.064 , $\hat{\theta} = 0.054 \pm 1.96 \times 0.036$; PPI-Qwen: rectifier -0.012 , $\hat{\theta} = 0.123 \pm 1.96 \times 0.039$
- (c) $n = 100$: coverage 94.7%, 94.3%, 95.4%, width 0.229, 0.204, 0.203 (labels, PPI-FinBERT, PPI-Qwen); gain for large N : 24% and 27%; at $n = 800$ PPI is wider (0.087 vs 0.081)
- Interpretation: the model mean converges to the wrong number; the labelled sample corrects it, and PPI is narrower only when $\text{Var}(f - Y) < \text{Var}(Y)$ and many texts are unlabelled

 MFM_ch15_valid_inference

B9 [Proposed]: Wild Cluster Bootstrap with 16 Stocks — Task

- **Model:** B2 [Solved]
- **Question:** with only 16 stock clusters, which of the nine slopes of B2 remain significant?
- **Data**
 - ▶ FNSPID panel of B2: 18 282 stock-days, 16 US stocks; score $z_{i,d}$ and excess return r^e as in B2
 - ▶ The nine regressions of B2 (three scores $\times h = 0, 1, 2$), SE clustered by stock ($G = 16$)
- **Tasks**
 - (a) Compute the p -value of each slope with t_{15} critical values.
 - (b) Run the wild cluster bootstrap with the null imposed [Cameron, Gelbach & Miller \(2008\)](#) ([What You Need \(20/20\)](#)), Rademacher $w_g = \pm 1$, $g(i)$: the stock of observation i , $\bar{y}$: the fit under H_0 (intercept only), 9 999 draws.
 - (c) Compute t with two-way clustering ([What You Need \(20/20\)](#)).
 - (d) Adjust the nine bootstrap p -values with Holm at 5%.
- **Report:** a table with t , the three p -values and the two-way t
- **Interpretation:** does Qwen at $h = 2$ survive?

B10 [Proposed]: Specification Curve for the Sentiment Strategy — Task

- **Model:** B3 [Solved] (a randomisation test)
- **Question:** over all reasonable variants of the strategy, is there more evidence of predictability than chance would give?
- **Data**
 - ▶ The sentiment strategy of A3 (What You Need (16/20)): position $\text{sign}(s_{i,d})$ from the close of $d + 1$, hedged with SPY, held one day; four trades on each day with positions; 3 568 trading days
 - ▶ Grid of Lecture 15: three scores $\times$ thresholds $|s| > b$, $b \in \{0, 0.25, 0.5\}$, $\times$ return day $d + 2, \dots, d + 5$: 36 strategies
- **Tasks**
 - (a) Compute the mean daily return and t_{NW} of each strategy and plot the sorted t -statistics Simonsohn, Simmons & Nelson (2020).
 - (b) Simulate the joint null: flip the signs of the positions in blocks of 10 signal days (dependent wild bootstrap Shao (2010)), the same flips for all 36, 2 000 times.
 - (c) Compute the joint p -values of the number of $t > 1.96$ and of the median t .
- **Report:** the sorted curve and the two joint p -values
- **Interpretation:** what would a "best specification" report hide?

A8 [Proposed]: The Qwen Strategy after Costs — Task

▣ **Model:** A3 [Solved]

▣ **Question:** the Qwen2.5-7B strategy has a statistically positive gross mean: is it tradable?

▣ **Inputs**

- ▶ The sentiment strategy of A3 (What You Need (16/20)): position $\text{sign}(s_{i,d})$ from the close of $d + 1$, hedged with SPY, held one day; four trades on each day with positions; 3 568 trading days
- ▶ Qwen2.5-7B signal: $\hat{\mu} = 5.64$ bp, $t_{NW} = 2.26$ ($L = 8$), $a = 0.950$, gross Sharpe ratio 0.63

▣ **Tasks**

- (a) Compute the break-even cost c^* .
- (b) Compute $\text{SE}(c^*)$ by the delta method and the 95% CI.
- (c) Recover σ and compute the net Sharpe ratio at 5 bp.

▣ **Report:** c^* , SE, CI, net Sharpe ratio

▣ **Interpretation:** does a significant gross mean imply a profit after costs?

B11 [Proposed]: Event Study with the Loughran–McDonald Tone — Task

▣ **Model:** B2 [Solved]

▣ **Questions**

- ▶ around negative and positive news days, when does the price move?
- ▶ does the dictionary separate the two groups as well as FinBERT?

▣ **Data**

- ▶ FNSPID panel of B2: 18 282 stock-days, 16 US stocks; score $z_{i,d}$ and excess return r^e as in B2
- ▶ The daily LM tone is exactly 0 on 48% of stock-days: groups $\text{tone} < 0$ (5 196 stock-days) and $\text{tone} > 0$ (4 288); FinBERT: bottom and top terciles

▣ **Tasks**

- For each date, average the excess returns of days $-5, \dots, +5$ over that date's stock-days in each group.
- Compute the mean over dates of the CAR for $[-5, -1]$, day 0, day 1 and $[+2, +5]$, with t_{NW} .
- Plot the mean cumulative paths of the four groups.

▣ **Report:** the table of the four windows and the chart

▣ **Interpretation:** why is the post-event return positive for both groups Tetlock (2007)?

C2 [Proposed]: Does Headline Sentiment Pay after Costs? — Task

▣ **Model:** [A3](#), [B2](#) [Solved]

▣ **Question:** does a daily strategy on FNSPID headline sentiment earn money after realistic costs?

▣ **Data**

- ▶ The sentiment strategy of [A3](#) ([What You Need \(16/20\)](#)): position $\text{sign}(s_{i,d})$ from the close of $d + 1$, hedged with SPY, held one day; four trades on each day with positions; 3 568 trading days
- ▶ Positions equally weighted; days without news: return 0; 5 bp per trade; scores FinBERT, Qwen2.5-7B, LM from the saved tables

▣ **Tasks**

- For FinBERT, compute the mean daily gross return, t_{NW} and a moving-block bootstrap 95% CI (blocks of 10 days, 2 000 draws).
- Compute the Sharpe ratio before and after costs, the break-even cost and the Sharpe ratio at 0, 1, 2, 5, 10 bp.
- Repeat for Qwen2.5-7B and LM, and for strong news $|s| > 0.5$ (exploratory); Holm over the three unfiltered strategies, or over all six if you select among them.
- Further step: correct the same-day slope with a random human-labelled sample of stock-days (divide by $\rho(s^*, \hat{s})$, or rectify the OLS moments as in [B8](#)).

▣ **Report:** a two-page note with a results table and charts, and a Quantlet folder

▣ **Interpretation:** discuss timing, look-ahead ([B3](#)), survivorship of the 16 stocks and crowding [Lopez-Lira & Tang \(2026\)](#)

C3 [Proposed]: Replicating Lopez-Lira and Tang on FNSPID — Task

▣ **Model:** C2, B2, B3, B9

▣ **Question:** does the published GPT-4 design Lopez-Lira & Tang (2026), run with an open model on the course headlines, find the same reaction and the same drift?

▣ **Inputs**

- ▶ Paper: GPT-4 answers YES, NO or UNKNOWN about each headline; $s = 1, -1, 0$ drives a daily long-short portfolio; overnight drift 0.34% per day, Sharpe ratio 2.97 (Table 1)
- ▶ Data: FNSPID headlines of the 16 stocks, adjusted daily closes of the course data; first safe return $d + 2$
- ▶ Model: Qwen2.5-7B-Instruct Qwen Team (2024), temperature 0, the prompt of Section 5 of the paper word for word

▣ **Tasks**

- Score every headline, parse the first line only, map YES/UNKNOWN/NO to 1/0/ - 1; report the shares and invalid answers.
- Average by stock-day and form the daily portfolio with the conventions of the next slide, earning the returns of day $d + 2$.
- Report the mean daily return in bp, t_{NW} , the hit rate, the Sharpe ratio, and the shares of days with one leg and with no position.
- Estimate Eq. (1) at headline level for $h = 0, 2$, with stock and date fixed effects, SE clustered by date and stock and the wild cluster bootstrap p (B9).
- Recompute the Sharpe ratio net of 5, 10, 20 bp per round trip (Figure 4) and compare with C2 on the same stock-days.

▣ **Report:** your numbers next to Tables 1 and 2 of the paper, cumulative returns before and after costs, a Quantlet folder

▣ **Interpretation**

- ▶ which differences from the paper can your data separate?
- ▶ what does B3 say about the bias?

C3 [Proposed]: Portfolio, Calendar and Cost Conventions

▣ Fixed before any run

- ▶ Weights (Table 1): long leg = equal-weighted mean return of the stocks with $s > 0$, short leg = same for $s < 0$, a leg only with at least two stocks; long-short = long minus short (gross exposure 2); if only one leg qualifies, hold it alone (footnote 20)
- ▶ Returns: raw stock returns, as in Table 1; the SPY-hedged version (as in C2) is a course adaptation, reported as a sensitivity: with both legs the hedge cancels, so it matters only on one-leg days
- ▶ Calendar: all trading days, as in C2, with return 0 without a position; mean, t_{NW} and Sharpe ratio over all days; hit rate over days with a position
- ▶ Costs (Figure 4): c bp per round trip on each unit of notional: $2c$ per day with two legs, c with one, plus, in the hedged version, c on the net SPY notional (1 on one-leg days)
- ▶ Regressions (d): raw returns in % per day, as in Table 2; the date fixed effects absorb the market return; the wild cluster bootstrap keeps both fixed effects in the null model

References (1/3)

- ▣ Aigner, D.J. (1973). Regression with a binary independent variable subject to errors of observation. *Journal of Econometrics*, 1(1), 49–59.
- ▣ Angelopoulos, A.N., Bates, S., Fannjiang, C., Jordan, M.I., Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671), 669–674.
- ▣ Araci, D. (2019). FinBERT: financial sentiment analysis with pre-trained language models. arXiv:1908.10063.
- ▣ Cameron, A.C., Gelbach, J.B., Miller, D.L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.
- ▣ Dawid, A.P., Skene, A.M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. *Applied Statistics*, 28(1), 20–28.
- ▣ DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.
- ▣ Dong, Z., Fan, X., Peng, Z. (2024). FNSPID: a comprehensive financial news dataset in time series. *Proceedings of the 30th ACM SIGKDD Conference*, 4918–4927.
- ▣ Fleiss, J.L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5), 378–382.

References (2/3)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Glasserman, P., Lin, C. (2023). *Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis*. arXiv:2309.17322.
- ▣ Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q. (2017). *On calibration of modern neural networks*. ICML; arXiv:1706.04599.
- ▣ Lopez-Lira, A., Tang, Y. (2026). *Can ChatGPT forecast stock price movements? Return predictability and large language models*. *Journal of Financial Economics*, 184, 104335.
- ▣ Loughran, T., McDonald, B. (2011). *When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks*. *Journal of Finance*, 66(1), 35–65.
- ▣ Malo, P., Sinha, A., Korhonen, P., Wallenius, J., Takala, P. (2014). *Good debt or bad debt: detecting semantic orientations in economic texts*. *Journal of the Association for Information Science and Technology*, 65(4), 782–796.
- ▣ McNemar, Q. (1947). *Note on the sampling error of the difference between correlated proportions or percentages*. *Psychometrika*, 12(2), 153–157.
- ▣ Newey, W.K., West, K.D. (1987). *A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix*. *Econometrica*, 55(3), 703–708.

References (3/3)

- ▣ Petersen, M.A. (2009). [Estimating standard errors in finance panel data sets: comparing approaches](#). *Review of Financial Studies*, 22(1), 435–480.
- ▣ Qwen Team (2024). [Qwen2.5 technical report](#). arXiv:2412.15115.
- ▣ Sarkar, S.K., Vafa, K. (2024). [Lookahead bias in pretrained language models](#). SSRN Working Paper 4754678.
- ▣ Shao, X. (2010). [The dependent wild bootstrap](#). *Journal of the American Statistical Association*, 105(489), 218–235.
- ▣ Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). [Specification curve analysis](#). *Nature Human Behaviour*, 4(11), 1208–1214.
- ▣ Tetlock, P.C. (2007). [Giving content to investor sentiment: the role of media in the stock market](#). *Journal of Finance*, 62(3), 1139–1168.
- ▣ Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M. (2020). [MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained Transformers](#). *NeurIPS*; arXiv:2002.10957.